• Solidification In AM:

(I) Equilibrium:

In the conventional casting or laser/electron-beam AM techniques, after melting, the liquid metal converts to a solid form through the cooling process known as solidification. Thermodynamically, solid and liquid equally have the same energy at their melting stage, thus, both are steady at the melting point. During the cooling process, the liquid metal starts to lose energy, which is the latent heat. Consequently, the liquid temperature is dropped, which minimizes the average interatomic space between movable and disordered atoms. On further cooling, the interaction forces of atoms stop them from moving away from each other and finally an absolute transformation of liquid to solid occurs. Transformation of a homogeneous liquid into a perfect homogeneous solid structure is enormously difficult to obtain in practice. It is important to study the redistribution of solute at the micron level, which will happen if formed solid crystal structures have different chemical compositions compared to the liquid. This can result in undesirable segregation or extreme refinement of solid, which must consider the situation under which the solid/liquid interface becomes unstable to form cellular or dendritic structures or even crystal imperfections. The free energy of a material is a function of pressure, temperature and composition. Equilibrium condition is obtained if the Gibb's free energy is at minimum level. Equilibrium does not exist in actual systems. The structure of a material which is a function of composition and temperature, can be identified from an equilibrium phase diagram using the assumption of a relatively slow transformation rate or a faster diffusion rate.

(II) Non- Equilibrium:

Since AM comprises rapid melting of metallic materials, knowledge of the theory of solidification plays an important role in predicting and monitoring the technique. The laser or EB based AM methods entail a localized moving heating source with a very short interaction time. Given the material's thermal conductivity, the solidification stage followed by the cooling duration may take twice the heating time. Hence, the solidification and cooling rates are very high, which are generally ascribed by the surrounding heat conduction/convection regimes, mainly due to the substrate material. As the point heat source delivers extremely focused energy, it causes vastly localized heat flux in the melt pool zone, together with a massive temperature gradient in the deposited layers. The temp gradient at the bottom of the substrate surface is higher compared to the top of the deposited surface.

In AM, the metallurgy is controlled by the chemistry of raw materials and the thermal history exposed to the material during the manufacturing technique. For different AM techniques, the heat transfer mechanism may differ, where a full melting occurs in laser or EB processed AM, they have the same metallurgical principles. Solidification followed by cooling to the ambient temperature controls the early phase distribution and grain morphology in the deposited layer. Moreover, the metal pool geometry is governed by the heat source speed, power and size, which consequently controls the solidification kinetics. It has become evident that the rapid solidifation in the melt pool offers the prospect of improving mechanical properties by allowing larger constitutional and structural latitude than is probable in the conventional manufacturing approach.

- Factors Affecting Solidification In AM:

(i) Cooling rate:

- The structure of the AM product is controlled by the thermal behaviour of the process, precisely by the temperature gradients, Solidification and cooling rates. They have multiple repeated heating and cooling cycles.
- Generally, cooling rates are influenced by heat input by the heab input, which is manipulated by the laser or EB power, beam scan speed layer thickness, Scanning strategies etc...
- When the laser power is low and scanning speed is high, this combination normally delivers a small heat flux that results in a larger cooling rate. On the contrary, with a higher laser power, and a lower scan speed, heat input would be intensified to melt a large Substrate area and eventually results in a slower cooling rate.

(II) Temperature Gradient and Solidification Rate:

- Through the solidification, columnar grains advance along the path of a higher temperature gradient in the melt pool.
- Various melt pool geometries impact the temperature gradient and eventually influence the grain structure in terms of type, size, texture and shape.
- The spherical melt pool generates curved and tapered columnar structures because of the deviation in the thermal gradient path from the pool border.
- However, the comet featured melt pool creates conventional and wide columnar structures, where the path of the maximum thermal gradient does not shift notably through the process.

(iii) Process Parameters:

- The solidified microstructure and the phase formations are considerably controlled by the input process parameters.
- Together with the higher specific energy and the faster deposition rate, the liquid melt will be at a higher temperature for an extended time to lower the temperature gradients. Therefore, grains are allowed to grow coarsened and mainly equiaxed.
- On the other hand, the minor specific energy is understood by applying a faster scan speed, and hence no adequate time for grain growth.
- When the specific energy is bought down, the energy required per unit area to melt down the powder is lowered. This calls for a need to lower the layer thickness.
  However, the thick layer causer lower cooling and results in a coarser grain structure.

(iv) Solidification Temperature Span:

- Usually, the broader solidification temperature span creates a greater solid/liquid or mashy zone, which is mostly responsible for the solidification cracking, as the liquid cannot allow load.
- This temperature range is altered by several factors, such as the existence of impurities like sulfur and phosphorous in ferrous or nickel alloys and some specific alloying elements to enhance mechanical properties.

(V) Gas Interactions:

- It is important to ensure appropriate gas environment to fabricate quality products. In AM techniques, argon and nitrogen are generally used ~~for~~ to offer an inert atmosphere and satisfy the high tolerance criteria.
- Eg: In EB-PBF, vacuum is required in the chamber.
  and, in laser DED, helium is used for shielding to enhance temperature temporal behaviour.